

XIAOYING(Camilla) XUE

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EDUCATION

University of Toronto

Toronto, Canada

- Honours Bachelor of Science
 - Computer Science Specialist, Major in Statistical Science, Economics Minor
 - cGPA: 3.94

2021 - Present

RESEARCH EXPERIENCE

Research Assistant

Toronto, Canada

- Professor Niv Dayan
 - Conducted comprehensive benchmarking of **RocksDB**, evaluating binary search and hash-based indexing to quantify performance improvements and identify read/write throughput bottlenecks
 - Designed, implemented, and assessed a minimal perfect hash index integrated within the SST data-block write and read pipelines to optimize index lookups and enhance read throughput

Fall 2025 - Present

Research Assistant

Toronto, Canada

- Professor Qizhen Zhang
 - Performed in-depth annotation of **PostgreSQL** source code to categorize client-server communication pathways, enabling systematic benchmarking of communication latency

Summer 2025 - Present

WORK EXPERIENCE

Software Engineer Intern

Toronto, Canada

- Quanta Technology
 - Contributed as a full-stack developer in an **Agile** environment and **Git** for version control to develop a finite-state-control workflow portal with **React** on the front end and **C#** on the back end, significantly reducing manual overhead and improving client operation accuracy.
 - Developed a web-based software tool for utility companies, with a focus on parsing and managing deeply nested tree-structured JSON data to support CRUD operations through **RESTful APIs**
 - Collaborated with cross-functional teams to design database solutions, implementing **PL/SQL** procedures and triggers to automate sequential business processes and enforce data integrity.

Summer 2024 – Fall 2025

RELEVANT EXPERIENCE

Pet Adoption Website

Toronto, Canada

- *Computer Science Team Project*
 - Planned and designed UI/UX workflows using **HTML**, **CSS**, and **JavaScript**, implemented a dynamic front-end with **React** to deliver a responsive user experience
 - Developed **RESTful** web services using **Django REST Framework**, extending back-end functionality to support **CRUD operations** across multiple user roles with custom **Python** code
 - Performed API quality assurance using **Postman** to test request methods, input validation, and access permissions, ensuring robust and secure back-end interactions

Fall 2023

TEACHING & OUTREACH

Teaching Assistance

Toronto, Canada

- STA130: An Introduction to Statistical Reasoning and Data Science
 - Hosted weekly 2-hour tutorials for a group of 25 students, assisted students' research projects.

Fall 2025, Winter 2024

Volunteer High school Mentor

Toronto, Canada

- WICS (Women in Computer Science)
 - Held regular one-on-one mentoring sessions with high school students to provide academic support.

Fall 2023

SKILLS

- **Programming Languages:** Bash, C, C++, C#, Java, TypeScript, JavaScript, Python, R, SQL, HTML, CSS
- **Frameworks & Libraries:** React, Bootstrap, Django, NumPy, Pandas
- **Languages:** English, Mandarin, Cantonese, Japanese